Woojung Song

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RESEARCH INTERESTS

I study how language models and agents behave in practice, with a focus on reliability and human simulation.

- **Agent behavior and reliability:** Characterizing task behavior (*AgentHabit*, preprint); improving tool use (*PA-Tool*, ACL 2026), evaluating tool trust (*Tool Overreliance*, preprint), and testing interactions with non-collaborative users (*NCUser*, ICLR 2026).
- **Values, personas, and human simulation:** Measuring values in context (*Value Portrait*, ACL 2025); testing whether self-reported traits predict behavior (*Psychometric Questionnaires*, EMNLP 2026); and evaluating character change over time (*ArcANE*, EMNLP 2026).

EDUCATION

Seoul National University, Seoul, South Korea — Sep 2025 – Present
Integrated M.S.–Ph.D. in Data Science
Advisor: Yohan Jo, HOLI Lab (Human-Oriented Language Intelligence)

Hanyang University, Seoul, South Korea — Mar 2022 – Aug 2025
B.I.S. in Information Systems — GPA: 4.35/4.50 (overall and major)

PEER-REVIEWED PUBLICATIONS

* Equal contribution. Includes accepted papers.

ArcANE: Do Role-Playing Language Agents Stay in Character at the Right Time?
EMNLP, Main Conference (accepted) 2026
Woojung Song*, Nalim Kim*, Sangjun Song, Chaewon Heo, Jongwon Lim, Yohan Jo

Human Psychometric Questionnaires Mischaracterize LLM Behavior
EMNLP, Main Conference (accepted) 2026
Woojung Song*, Dongmin Choi*, Yoonah Park, Jongwook Han, Eun-Ju Lee, Yohan Jo

Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models
ACL, Main Conference 2026
Jonggeun Lee*, **Woojung Song***, Jongwook Han, Haesung Pyun, Yohan Jo

Non-Collaborative User Simulators for Tool Agents
International Conference on Learning Representations (ICLR) 2026
Jeonghoon Shim, **Woojung Song**, Cheyon Jin, Seungwon Kook, Yohan Jo

Psychometric Item Validation Using Virtual Respondents with Trait-Response Mediators
Transactions of the Association for Computational Linguistics (TACL) 2026
Sungjib Lim, **Woojung Song**, Eun-Ju Lee, Yohan Jo

Quantifying Data Contamination in Psychometric Evaluations of LLMs
Findings of EACL 2026
Jongwook Han*, **Woojung Song***, Jonggeun Lee*, Yohan Jo

Interpretable Prediction of Private Brand Purchases by Pet Type in E-Commerce for Consumer Behavior Analysis Using Real-World Transaction Data
PeerJ Computer Science 2026
Jaehyuk Lee*, **Woojung Song***, Jina Kim, Yoona Chung, Eunchan Kim

Value Portrait: Assessing Language Models' Values through Psychometrically and Ecologically Valid Items
ACL, Main Conference 2025
Jongwook Han*, Dongmin Choi*, **Woojung Song***, Eun-Ju Lee, Yohan Jo

PREPRINTS

* Equal contribution. All papers below are under review.

AgentHabit: Characterizing Distinct Behaviors of Agents on Everyday Tasks
arXiv:2609.32795 2026
Woojung Song*, Hoyeol Yang*, Jeonghoon Shim, Sungjib Lim, Jonggeun Lee, Yunho Choi, Yohan Jo

Agents' Overreliance on Unreliable Tools
arXiv:2609.05587 2026
Hoyeol Yang*, **Woojung Song***, Taewon Kim, Jonghyun Song, Seoyeon Park, Yohan Jo

Source Preference in the Wild: How LLM Agents Favor Items by Source, and How to Reduce It
arXiv:2610.03195 2026
Jonghyun Song*, Haewon Park*, Jeonghoon Shim*, **Woojung Song**, Yohan Jo

RESEARCH EXPERIENCE

Seoul National University, HOLI Lab — Seoul, South Korea
Graduate Research (Integrated M.S.–Ph.D.) — Sep 2025 – Present

- Co-first-authored AgentHabit: a benchmark profiling 18 models across 23 behavioral axes on 86 everyday tasks.
- Co-first-authored studies of tool overreliance, temporal character consistency (ArcANE), and the behavioral validity of human psychometric questionnaires.

Seoul National University, HOLI Lab — Seoul, South Korea
Undergraduate Research Intern — Jul 2024 – Aug 2025

WORK EXPERIENCE

LG Electronics, H&A Sound/Image Intelligence R&D Lab — Seoul, South Korea
AI Research Intern — Mar 2024 – Jun 2024

- Improved LLM-based web question answering using retrieval-augmented generation and multi-agent approaches.
- Developed WAAG, an automated QA evaluation system with a 40% improvement in accuracy.
- Achieved the highest level in the LG Software Competency Certification Test.

Republic of Korea Army, KATUSA, U.S. Eighth Army — Pyeongtaek, South Korea
Sergeant (Senior KATUSA) — May 2020 – Nov 2021

- Completed mandatory military service as a Korean Augmentation to the United States Army (KATUSA) soldier.
- Received a Certificate of Appreciation for exceptional service.

SCHOLARSHIPS

Presidential Science Scholarship for Graduate Students — South Korea
Ministry of Science and ICT — 2026 – 2027